MGMT 405 – Managerial Economics

Teaching Note – Module 3: Deriving the Bang-for-the-Buck Rule

This note derives the rule for the optimal mix of capital and labor in the **long run**, when both inputs can be adjusted.

Marginal analysis requires that for the optimal use of a flexible input, its marginal benefit must equal its marginal cost (MB=MC). In the context of inputs in production, the marginal benefit is "by how much do our revenues increase if we use more of this input?" This is the marginal revenue product (*MRP*) of this input. The marginal cost is: "By how much do our costs go up if we use more of this input?" In the case of workers, this is the wage (w); in the case of capital, it is the price of capital (p_K).

Note that we assume constant input prices, so we do not need to worry about w going up if we hire more workers, or p_K going up (or down) if we buy more machines. We thus have the following two formulas for optimization:

For labor: $\mathrm{MRP}_L = w$

For capital: $\mathrm{MRP}_K = p_K$

Consequently,

$$\frac{\mathrm{MRP}_L}{\mathrm{MRP}_K} = \frac{w}{p_K}$$

Recall from the Teaching Note on Hiring Decisions in the Short Run that $\mathrm{MRP}_L = \mathrm{MR} \cdot \mathrm{MP}_L$, i.e., the product of marginal revenue and the marginal product of labor. Similarly, for capital we have: $\mathrm{MRP}_K = \mathrm{MR} \cdot \mathrm{MP}_K$. For example, if we use one more unit of capital (e.g., a robot), then output goes up by MP_K units, and each extra unit of output, in turn, raises our revenues by MR. Thus, MRP_K tells us by how much our revenues increase if we use one more unit of capital. Replacing these two in the equation above yields:

$$\frac{\mathrm{MR} \cdot \mathrm{MP}_L}{\mathrm{MR} \cdot \mathrm{MP}_K} = \frac{w}{p_K}$$

Now MR cancels out. This leaves:

$$\frac{\mathrm{MP}_L}{\mathrm{MP}_K} = \frac{w}{p_K}$$

Re-arranging then yields the **bang-for-the-buck rule**:

$$\frac{\mathrm{MP}_L}{w} = \frac{\mathrm{MP}_K}{p_K}$$

Three conditions come with the rule. It holds in the long run, when both labor and capital are flexible. It refers to a given output quantity Q: it tells us the cheapest input mix for producing that quantity, not how much to produce. And it assumes the input prices w and p_K are constant.